

LAMDAS, CURRYING, AND ENVIRONMENTS

COMPUTER SCIENCE MENTORS 61A

September 14, 2026 - September 18, 2026

1 Expressions and Truth Values

```
1. def quick_maths(man, free):  
    print(man)  
    if man % 2 == 0 and free:  
        return quick_maths(man + 2, not free)  
    else:  
        man -= 1  
        if man == 3:  
            print(man)  
        else:  
            return man  
big_shaq = 2
```

Expression	Interactive Output
<code>print(quick_maths(big_shaq, True))</code>	

2 Lambda Expressions and Function Forms

1. What do lambda expressions do?

2. Rewrite each function in the requested equivalent form. Preserve the function's parameters and behavior; only change whether it is written with **def** or **lambda**. For your reference, a **lambda** assignment assigns a lambda function to a variable.

(a) Rewrite the following **def** as a lambda assignment.

```
def square(x):  
    return x * x
```

(b) Rewrite the following lambda assignment using **def**.

```
subtract = lambda x, y: x - y
```

(c) Rewrite this zero-argument function as a lambda assignment.

```
def greeting():  
    return 'Hello, world!'
```

3 Currying and Zero-Argument Functions

1. Fill in the blanks (*without using any numbers in the first blank*) such that the entire expression evaluates to 9.

```
(lambda x: lambda y: _____) (_____) (lambda z: z*z) ()
```

4 Environment Diagrams with Lambdas

1. Give the environment diagram and console output that result from running the following code.

```
x = 20
def foo(y):
    x = 5
    def bar():
        return lambda y: x - y
    return bar

y = foo(7)
z = y()
print(z(2))
```

2. Draw the environment diagram that results from running the code below.

```
def dream1(f):
    def dream2(secret):
        mind = f(secret)
        kick = lambda x: mind()
        return kick(2)
    return dream2

inception = lambda secret: lambda: secret
real = dream1(inception)(42)
```

5 Basic Function Implementation

1. Write a higher-order function that passes the following doctests.

Challenge: Write the function body in one line.

```
def mystery(f, x):  
    """  
    >>> from operator import add, mul  
    >>> a = mystery(add, 3)  
    >>> a(4) # add(3, 4)  
    7  
    >>> a(12)  
    15  
    >>> b = mystery(mul, 5)  
    >>> b(7) # mul(5, 7)  
    35  
    >>> b(1)  
    5  
    >>> c = mystery(lambda x, y: x * x + y, 4)  
    >>> c(5)  
    21  
    >>> c(7)  
    23  
    """
```

2. Implement `make_skipper`, which takes in a number `n` and outputs a function. When this function takes in a number `x`, it prints out all the numbers between 0 and `x`, skipping every `n`th number.

```
def make_skipper(n):  
    """  
    >>> a = make_skipper(2)  
    >>> a(5)  
    1  
    3  
    5  
    """
```